

Instructor

Prof. Lea Hildebrandt Ruiz

Contact info lh@che.utexas.eduOffice hours Tue 1-2pm, CPE 5.458; Wed 10.30 am – 11.30 am on Zoom
or by appointment**Teaching Assistant**

Shihao Zhai

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Office hours Mo 5-6pm; Wed after recitation; Thu 5-6pm, locations TBD

Responsibilities recitation, office hours to assist with HW & conceptual problems,
assist with exam grading**Tutor**

Minseong Jung

Contact info minseongjung05@gmail.com

Office hours Mo and Fri 3.30-5pm, locations TBD

Responsibilities office hours to assist with HW & conceptual problems

Grader

Arya Balasubramanian

Contact info aryamba2004@gmail.com

Responsibilities grade HW

Lecture

TTh 9:30 am – 10:45 am CPE 2.220

NOTE: HW is generally due at **9.15 am** on Tuesdays.**Recitation**

W 3:00 – 5:00 pm ASE 1.126

NOTE: Midterm exams will be conducted during recitation.

Attendance is strongly suggested as there will be unannounced quizzes in lecture and recitations. Most lectures (and some quizzes) will be based on reading assignments that you should complete prior to coming to class.

Text

Elementary Principles of Chemical Processes, by Felder, Rousseau and Bullard, 4th edition, Wiley Press.

<https://www.wiley.com/en-us/Elementary+Principles+of+Chemical+Processes%2C+4th+Edition-p-9781119192107>

Course Website: A course page will be maintained on Canvas—Please check it regularly!

Grading:

Anticipated course grade cut-offs are A: 93 %, A-: 90%, B+:87%, B:83 %, B-:80%, C+:77%, C: 73%, C-:70%, D+: 67%, D:63%, D-: 60%. Grades may be curved, i.e. these grade cut-offs may be lowered; they will not be raised.

Two grading schemes will be offered this semester. You will choose the grading scheme on the last day of class. There will be 4 midterm exams.

Scheme A

Homework	10%
Best 4 of 5 Quizzes	20%
Exams	70%
Final Exam	0%

Scheme B

Homework	10%
Best 4 of 5 Quizzes	20%
Exams (Best 3 of 4)	52.5%
Final Exam	17.5%

Course Description and Objectives:

An introduction to concepts that are necessary to design and analyze industry-scale chemical processes. This course focuses on the most fundamental and wide-reaching aspects of chemical engineering – material and energy conservation in open and closed systems, both at steady state and in transient operation. The basics of thermodynamics and kinetics in the context of reactor, separator, and heat exchanger design are presented. Overall, this course emphasizes the principles and calculation techniques used in the field of Chemical Engineering, which form the foundation for all subsequent Chemical Engineering courses and analysis. Problem formulation and problem solving are stressed to equip students with life-long skills necessary for engineering and engineering-related careers.

Course Prerequisites:

- One year of college-level mathematics (algebra, solutions of simultaneous equations, quadratic equations, basic differentiation and integration, trigonometry and basic geometry) – through *M408D*
- Chemistry (stoichiometry, concept of mole, molecular weight, concentration, molarity, molality, etc.; enthalpy of reaction, ideal gas law, pressure, absolute temperature) – *Ch302*
- Basic knowledge of statics and dynamics in physics (concepts of force, work, pressure, kinetic energy, potential energy) – *Phy303* or high school physics
- Spreadsheet calculations for solving systems of linear equations, function minimization, and plotting – *ChE210*
- Time management skills to balance course load and assignments

Course procedure

All students are expected to attend and participate in lecture and recitation.

- Lecture will be used to introduce new material, demonstrate and work through detailed example problems.
- Recitation will be used to review difficult topics, work through additional example problems.
- Canvas will be used extensively in this course. Assignments will be posted on Canvas, as well as course materials and grades.

Reading assignments are required and you are responsible for the concepts and examples contained therein, even if not all content is covered in lecture or recitation. Complete the reading before the relevant class period.

In-class participation. All students are strongly encouraged to participate in lecture and recitation to maximize your learning.

Homework problems are for you to *practice* using the concepts and equations and are representative of those on exams. It is in your best interests to *struggle a bit* with the problems before seeking help, to understand why a certain approach is chosen and the details of each step. Homework submitted within 24 hours of the due date/time will be accepted with a 25 (out of 100) point penalty. Homework submitted more than 24 hours after the deadline will not be accepted. Exceptions can be made in the case of an excused absence – contact the instructor as soon as possible.

Quizzes will be administered in lecture or recitation. **There will be no make-up quizzes, but one of the quiz grades will be dropped.**

Exams will be based upon material covered in lecture, recitation, reading and in the homework.

- You are allowed to prepare and bring a “cheat sheet” for the exam on a 5×7 notecard. Required equations and tables will be provided in the exam – use the “cheat sheet” to summarize other key information.
- Clearly identify all balances, assumptions and any algebraic symbols used. Show all units and all algebraic and other mathematical calculations. Box your answers & be mindful of units and significant figures.
- Exams will be a combination of simpler problems (i.e., knowledge-based or those similar to HW and class examples) and more challenging problems (i.e., in which you apply the concepts in a slightly different way). These are intended to probe the depths of your understanding.

There will be no make-up exams, but grading scheme 2 allows for one exam drop.

Midterm exams are scheduled for **September 17, October 8, October 29 and November 19**. The final exam is scheduled by the university, currently for Monday **December 15, 10:30 am-12:30 pm**.

Regrade requests will be accepted up to one week after the return of the homework, quiz or exam and must be accompanied by a written note explaining why a different grade is deemed appropriate. Do not write on your graded homework/exam. A re-grade request may trigger a re-grade of the entire assignment, and your exam may be compared to your exam before it was returned. Submit your HW regrade requests via Gradescope.

Absences

If you are ill or experience another personal emergency on a day of a quiz or exam, obtain official documentation for that emergency (e.g. a letter from student health services, student emergency services, etc.). **If officially documented, your absence will be excused. Please do not come to class when you are ill.** Similarly, if you will be absent due to observance of a religious holy day, by UT Austin policy, notify the instructor of your pending absence at least fourteen days prior to the date of observance of a religious holy day. In this case, your absence will be excused.

Statement on Flexibility

This course will reaffirm one of the core values here at UT Austin: responsibility. Our responsibility to ourselves and each other is to put our humanity in the forefront of our academic pursuits. If you experience any hardships such as illness, accident, family crisis please know that these policies may be amended and therefore you should communicate with me as soon as you feel comfortable doing so. If for any reason you do not feel comfortable discussing with me, please visit Student Emergency Services.

Feedback

It is my goal to make this an excellent course and a positive experience for all of us. If you feel that the course is not meeting your expectations or you want to provide feedback on how the course is progressing for you, please contact me and / or the teaching assistant. I will also be circulating an early course evaluation in September. I will use your constructive comments as a way to improve the course where possible.

How to succeed in this class

- Diagram the problem - The math is not as difficult as envisioning the physical situation – always draw pictures to identify the direction of mass transfer, flow, etc.
- Do the homework – preferably not just the night before it is due and try to understand the steps. You are allowed to work in groups but need to understand all of the work that you turn in – including problem set up, calculations and solutions.
- Do more than the HW – read the book and example problems. (I on purpose do not discuss most example problems from the book during class so that you have additional problems to practice with.) After each lecture, look over and try to understand your notes and try to work in-class example problems on your own. If something is unclear, ask us!
- Ask questions – help us help you! I will be collecting your feedback after lecture, asking you about your level of understanding of the topics we discussed. **I will use input from this survey to cover a brief review at the beginning of the following lecture.** Thus, please be sure to fill out the survey if anything is unclear. We also have plenty of office hours scheduled each week and can also meet with you by appointment.

Knowledge, abilities and skills students should gain from this course:

- Exposure to systems analysis and the concept of unit operations.
- Exposure to library and reference data sources and the need to use them as an engineer.
- Ability to make calculations using different sets of units, including the appropriate use of the gravitational conversion factor and gauge vs. absolute pressure.
- Ability to write and balance chemical reaction equations. Ability to identify the excess and limiting reactants and calculate the percent conversion and yield of a reaction.
- Ability to define steady-state and distinguish between open and closed systems.
- Ability to solve material balances for systems with or without a chemical reaction. Ability to solve a number of interconnected units involving recycle, bypass and purge.

- Ability to define vapor pressure, saturation, equilibrium, triple point, dew point, partial saturation (humidity), etc.
- Ability to apply Raoult's and Henry's laws for multi-component systems.
- Ability to define heat, work, energy, enthalpy, etc.
- Ability to make estimations of heat capacity and to calculate enthalpy changes, with and without phase changes.
- Ability to use the steam tables.
- Ability to solve energy balances for closed and open systems, with and without chemical reaction.
- Ability to solve simple combined material and energy balances (with and without chemical reactions).
- Ability to distinguish and make calculations for adiabatic and non-adiabatic processes.
- Ability to apply spreadsheet calculations to mass/ energy problems.

ABET Student Outcomes for this course:

This course addresses the following ABET student outcomes:

Student Outcome 1: Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

Impact on Subsequent Courses in Curriculum:

Virtually every subsequent ChE course requires material and energy balance concepts. In addition, properties of materials (gas laws, vapor pressures, etc.) are important in *ChE319*, *ChE322*, *ChE354*, *ChE350*, *ChE 363*, and *ChE473K*.

Academic Integrity:

The University has a strict policy on academic integrity. Any form of plagiarism or other misconduct will not be tolerated in ChE 317 and will be referred to Student Conduct and Academic Integrity in the Office of the Dean of Students. Please review the website below for more information, and feel free to contact the instructor with questions or concerns.

<https://deanofstudents.utexas.edu/conduct/>

Sharing of Course Materials is Prohibited:

No materials used in this class, including, but not limited to, lecture notes, videos, assessments (quizzes, exams, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code. Sites used for sharing materials are frequently monitored, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students.

Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form.

Classroom Accessibility and Inclusivity:

I am committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). Please refer to D&A's website for contact and more information: <https://disability.utexas.edu/> If you are already registered with D&A, please discuss your accommodations and needs with me as early as possible in the semester.

Important Safety Information:

If you have concerns about the safety or behavior of fellow students, TAs or Professors, call BCAL (the Behavior Concerns Advice Line): 512-232-5050. Your call can be anonymous. If something doesn't feel right – it probably isn't. Trust your instincts and share your concerns.

Title IX Reporting:

Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working environment free from discrimination in all its forms. When sexual misconduct occurs in our community, the university can:

1. Intervene to prevent harmful behavior from continuing or escalating.
2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation.
3. Investigate and discipline violations of the university's relevant policies. (<https://titleix.utexas.edu/policies>).

Beginning January 1, 2020, Texas Senate Bill 212 requires all employees of Texas universities, including faculty, report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. **I am a Responsible Employee and must report any Title IX related incidents** that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a responsible employee. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/>, contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.

Live class schedule and reading assignments

The content is organized into weekly modules. While the readings are divided by class meeting, the lecture/recitation material may shift around to allow flexibility.

This is our course plan as of Aug 29, 2025. The details of this plan may need to change; any changes will be communicated clearly via a course announcement and updated syllabus.

WEEK/MODULE 1		
8/26 TUE	Welcome, course overview ChE 210 review Introduction to material balances: single, non-reactive units and multi-unit systems without reaction. DOF analysis.	Syllabus
8/27 WED		Ch. 1-3
8/28 THU		4.1-4.4, DoF handout
WEEK/MODULE 2:		
9/2 TUE	MB with recycle, bypass and purge Reaction terminology: limiting reagents, stoichiometry, yield, conversion. MB with reactions	4.5
9/3 WED		
9/4 THU		4.6
WEEK/MODULE 3		
9/9 TUE	Multi-unit processes with reactions	4.6-4.8
9/10 WED		
9/11 THU		
WEEK/MODULE 4		
9/16 TUE	Exam 1 Ideal gases and mixtures with applications to MB	
9/17 WED		
9/18 THU		5.1-5.2
WEEK/MODULE 5		
9/23 TUE	Real gases & mixtures	5.3-5.4
9/24 WED		
9/25 THU		
WEEK/MODULE 6		
9/30 TUE	Single component phase equilibria	6.1-6.3
10/1 WED		
10/2 THU		
WEEK/MODULE 7		
10/7 TUE	Vapor pressure: Antoine eqn, Cox chart, steam tables Exam 2 Solid/liquid and liquid/liquid equilibria	6.1-6.4
10/8 WED		
10/9 THU		6.5-6.6
WEEK/MODULE 8		
10/14 TUE	Adsorption, Review	6.7
10/15 WED		
10/16 THU	Energy types, EB on closed systems	7.1-7.3
WEEK/MODULE 9		

10/21 TUE	EB on open systems, Mechanical EB	7.4-7.6
10/22 WED		
10/23 THU		
WEEK/MODULE 10		
10/28 TUE	Calculating H and U when not tabulated	8.1-8.3
10/29 WED	Exam 3 enthalpy for phase change, psychometric charts	
10/30 THU		8.4-8.5
WEEK/MODULE 11		
11/4 TUE	enthalpy of mixing, heat of reaction	8.5, 9.1-9.2
11/5 WED		
11/6 THU	Reactive energy balances	9.3-9.5
WEEK/MODULE 12		
11/11 TUE	Fuels and combustion	9.6
11/12 WED		
11/13 THU		
WEEK/MODULE 13		
11/18 TUE	Transient balances Exam 4	10.1-10.2
11/19 WED		
11/20 THU		
WEEK/MODULE 14		
12/2 TUE	Transient balances	10.3
12/3 WED		
12/4 THU	Transient coupled MB and EB	10.4
	CUMULATIVE FINAL EXAM	